

SOHAM DAS

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EDUCATION

University of Cincinnati, Cincinnati, Ohio
Bachelor of Science, Computer Science

Graduating: May 2027

GPA: 3.9

- Primary Coursework: Artificial Intelligence, Linear Algebra, Probabilistic Models, Data Structures & Algorithms, Software Engineering

SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, Java, SQL

Machine Learning: PyTorch, Computer Vision, Object Detection, 3D Perception, Model Quantization, ONNX

Infrastructure: Docker, CUDA, CARLA Simulator, Git, PostgreSQL, REST APIs

Specializations: Autonomous Systems, Performance Optimization

EXPERIENCE

Autonomous Driving Researcher | ACSI Laboratory, Feng Chia University, Taiwan Feb 2026 – Aug 2026

- Trained a **YOLOv4** traffic sign detector to **95% mAP@0.5** and deployed it to an **Orange Pi 5 Ultra** NPU at **60-70 FPS** via a PyTorch → ONNX → RKNN quantization pipeline
- Reproduced the **SparseOcc 3D occupancy model** on nuScenes (0.368 RayIoU, matching published benchmark) in a from-scratch reproducible **Docker environment**
- Resolved a scenario-resolution bug in the CARLA-based Fail2Drive benchmark causing silent test-scenario skips; fix merged upstream

Backend Developer Intern | Wizzer

May 2025 – Aug 2025

- Optimized **REST API performance**, reducing response times by **40–60%** via query optimization, N+1 elimination, and asynchronous execution
- Established 8-week **API latency monitoring** across **P50–P99**; identified and flagged an upstream data-throttling artifact that would otherwise have been misread as a performance gain
- Engineered a **mobile user acquisition system** with deep link fallbacks and dynamic QR code routing, achieving **100% error-free redirects** and device-based app store detection

Lab Grader | College of Engineering and Applied Science, University of Cincinnati

Sep 2023 – Dec 2023

- Evaluated 10 C++ programming labs for freshman-level course, focusing on **object-oriented programming** and **algorithm design**
- Provided feedback for **26 students** on documentation, coding efficiency and logical accuracy

RESEARCH

Privileged Occupancy Filtering for Fixed-Vocabulary End-to-End Driving

Apr 2026 – Aug 2026

- Designed a **1,665-job closed-loop intervention study** (185 routes × 3 seeds × 3 arms) testing whether privileged **occupancy filtering** reduces collisions in the **SparseDriveV2** planner on the **Fail2Drive** benchmark, using process isolation and permutation-balanced arm ordering to eliminate confounds
- Found a **4.0-point reduction** in collision-route incidence (95% CI [-7.7, -0.4]), consistent in sign across all three seeds, satisfying a pre-specified route-completion guardrail; Driving Score and Success Rate also improved significantly

AWARDS & HONORS

Taiwan Experience Education Program (TEEP) Scholar

Feb 2026 – Aug 2026

International Collegiate Programming Contest (ICPC) Competitor for UC

Nov 2024

International Outreach Scholarship

Aug 2022 – Present

Dean's List

Aug 2022 – Present